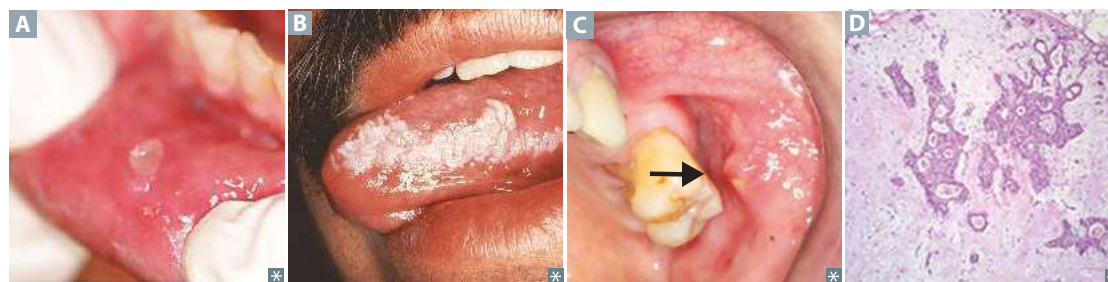


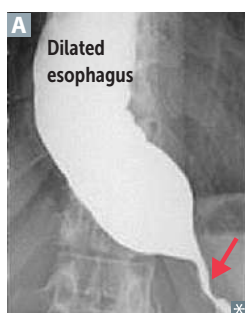
► GASTROINTESTINAL—PATHOLOGY

Oral pathologies

Aphthous ulcers	Also called canker sores. Common oral lesions that appear as painful, shallow, round to oval ulcers covered by yellowish exudate A . Recurrent aphthous stomatitis is associated with celiac disease, IBD, SLE, Behçet syndrome, HIV infection.
Squamous cell carcinoma	Most common malignancy of oral cavity. Usually affects the tongue. Associated with tobacco, alcohol, HPV-16. Presents as nonhealing ulcer with irregular margins and raised borders. Leukoplakia (white patch B) and erythroplakia (red patch) are precursor lesions.
Sialolithiasis	Stone formation in major salivary gland ducts (parotid C , submandibular, or sublingual). Associated with salivary stasis (eg, dehydration) and trauma. Presents as recurrent pre-/periprandial pain and swelling in affected gland.
Sialadenitis	Inflammation of salivary gland due to obstruction, infection (eg, <i>S aureus</i> , mumps virus), or immune-mediated mechanisms (eg, Sjögren syndrome).
Salivary gland tumors	Usually benign and most commonly affect the parotid gland. Submandibular, sublingual, and minor salivary gland tumors are more likely to be malignant. Typically present as painless mass/swelling. Facial paralysis or pain suggests malignant involvement. ▪ Pleomorphic adenoma (benign mixed tumor)—most common salivary gland tumor D. Composed of chondromyxoid stroma and epithelium and recurs if incompletely excised or ruptured intraoperatively. May undergo malignant transformation. ▪ Warthin tumor (papillary cystadenoma lymphomatosum)—benign cystic tumor with germinal centers. May be bilateral or multifocal. Typically found in people who smoke. “Warriors from Germany love smoking.” ▪ Mucoepidermoid carcinoma—most common malignant tumor. Mucinous and squamous components.



Achalasia



Failure of LES to relax due to degeneration of inhibitory neurons (containing NO and VIP) in the myenteric (Auerbach) plexus of esophageal wall.

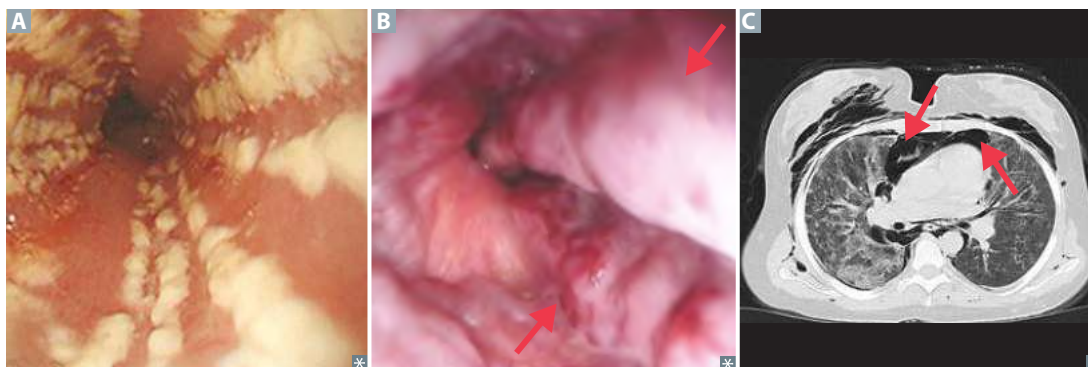
1° idiopathic. 2° from Chagas disease (*T cruzi* infection) or extraesophageal malignancies (mass effect or paraneoplastic). **Chagas** disease can cause **achalasia**.

Presents with progressive dysphagia to solids and liquids (vs obstruction—primarily solids). Associated with ↑ risk of esophageal cancer.

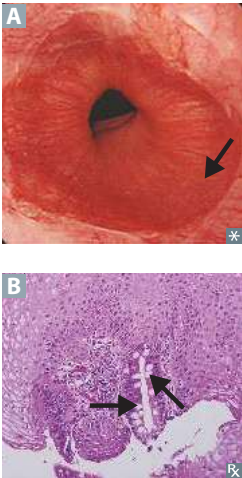
Manometry findings include uncoordinated or absent peristalsis with ↑ LES resting pressure. Barium swallow shows dilated esophagus with area of distal stenosis (“bird’s beak” **A**). Treatment: surgery, endoscopic procedures (eg, botulinum toxin injection).

Other esophageal pathologies

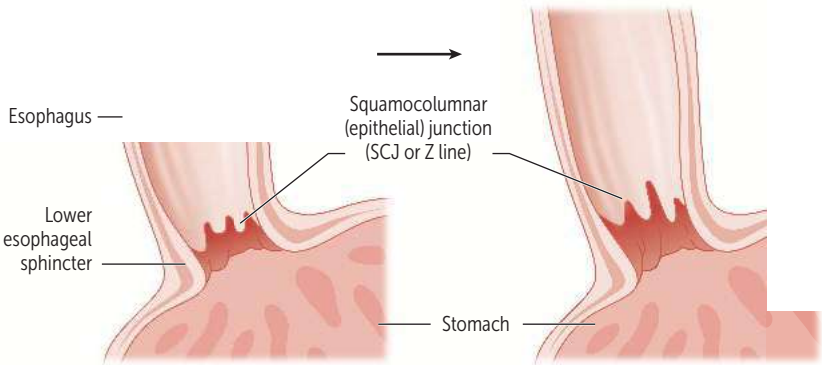
Gastroesophageal reflux disease	Transient decreases in LES tone. Commonly presents as heartburn, regurgitation, dysphagia. May also present as chronic cough, hoarseness (laryngopharyngeal reflux). Associated with asthma. Complications include erosive esophagitis, strictures, and Barrett esophagus.
Esophagitis	Inflammation of esophageal mucosa. Presents with odynophagia and/or dysphagia. Types: ▪ Reflux (erosive) esophagitis—most common type. 2° to GERD. ▪ Medication-induced esophagitis—2° to bisphosphonates, tetracyclines, NSAIDs, ferrous sulfate, potassium chloride. ▪ Eosinophilic esophagitis—chronic, immune-mediated, eosinophil-predominant. Associated with atopic disorders (eg, asthma). Esophageal rings and linear furrows on endoscopy. ▪ Infectious esophagitis—<i>Candida</i> (most common; white pseudomembranes A), HSV-1 (punched-out ulcers), CMV (linear ulcers). Associated with immunosuppression. ▪ Corrosive esophagitis—2° to caustic ingestion.
Plummer-Vinson syndrome	Triad of d ysphagia, i ron deficiency anemia, e sophageal webs. ↑ risk of esophageal s quamous cell carcinoma (" Plumber dies "). May be associated with glossitis.
Mallory-Weiss syndrome	Partial thickness, l ongitudinal l acerations of gastroesophageal junction, confined to mucosa/submucosa, due to severe vomiting. Often presents with hematemesis +/- abdominal/back pain. Usually found in patients with alcohol use disorder, bulimia nervosa.
Esophageal varices	Dilated submucosal veins (red arrows in B) in lower 1/3 of esophagus 2° to portal hypertension. Common in patients with cirrhosis, may be source of life-threatening hematemesis.
Distal esophageal spasm	Formerly called diffuse esophageal spasm. Spontaneous, nonperistaltic (uncoordinated) contractions of the esophagus with normal LES pressure. Presents with dysphagia and anginalike chest pain. Barium swallow may reveal “corkscrew” esophagus. Manometry is diagnostic. Treatment includes nitrates and CCBs.
Scleroderma esophageal involvement	Esophageal smooth muscle atrophy and fibrosis → ↓ LES pressure and distal esophageal dysmotility → acid reflux and dysphagia → stricture, Barrett esophagus, and aspiration. Part of CREST syndrome.
Esophageal perforation	Most commonly iatrogenic following esophageal instrumentation. Noniatrogenic causes include spontaneous rupture, foreign body ingestion, trauma, malignancy. Pneumomediastinum (arrows in C) and subcutaneous emphysema (signs include crepitus in the neck region or chest wall) can indicate dissecting air. Boerhaave syndrome —transmural, usually distal esophageal rupture due to violent retching.



Barrett esophagus



Specialized intestinal metaplasia (arrow in **A**)—replacement of nonkeratinized stratified squamous epithelium with intestinal epithelium (nonciliated columnar with goblet cells [arrows in **B**]) in distal esophagus. Due to chronic gastroesophageal reflux disease (GERD). Associated with ↑ risk of esophageal adenocarcinoma.



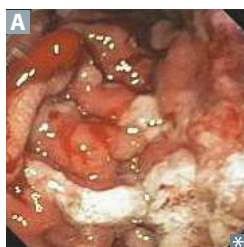
Esophageal cancer

Typically presents with progressive dysphagia (first solids, then liquids) and weight loss. Aggressive course due to lack of serosa in esophageal wall, allowing rapid extension. Poor prognosis due to advanced disease at presentation.

CANCER	PART OF ESOPHAGUS AFFECTED	RISK FACTORS	PREVALENCE
Squamous cell carcinoma	Upper 2/3	Alcohol, hot liquids, caustic strictures, smoking, achalasia, nitrosamine-rich foods	More common worldwide
Adenocarcinoma	Lower 1/3	Chronic GERD, Barrett esophagus, obesity, tobacco smoking	More common in America

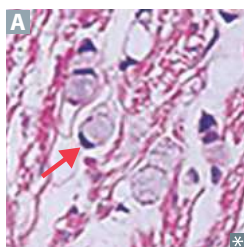
Gastritis

Acute gastritis	Erosions can be caused by: ▪ NSAIDs—↓ PGE₂ → ↓ gastric mucosa protection ▪ Burns (Curling ulcer)—hypovolemia → mucosal ischemia ▪ Brain injury (Cushing ulcer)—↑ vagal stimulation → ↑ ACh → ↑ H⁺ production	Especially common among patients with alcohol use disorder and those taking daily NSAIDs (eg, for rheumatoid arthritis) Burned by the Curling iron Always Cushion the brain
Chronic gastritis	Mucosal inflammation, often leading to atrophy (hypochlorhydria → hypergastrinemia) and intestinal metaplasia (↑ risk of gastric cancers)	
<i>H pylori</i>	Most common. ↑ risk of peptic ulcer disease, MALT lymphoma	Affects antrum first and spreads to body of stomach
Autoimmune	Autoantibodies (T-cell induced) to the H ⁺ /K ⁺ -ATPase on parietal cells and to intrinsic factor. ↑ risk of pernicious anemia	Affects body/fundus of stomach

Ménétrier disease

Hyperplasia of gastric mucosa → hypertrophied rugae (“**wavy**” like brain gyri **A**). Causes excess mucus production with resultant protein loss and parietal cell atrophy with ↓ acid production. Precancerous.

Presents with **W**eight loss, **A**norexia, **V**omiting, **E**pigastric pain, **E**dema (due to protein loss; pronounce “**WAVEE**”).

Gastric cancer

Most commonly gastric adenocarcinoma; lymphoma, GI stromal tumor (common mutations include *KIT* or *PDGFRA*), carcinoid (rare). Early aggressive local spread with node/liver metastases. Often presents late, with **W**eight loss, **E**arly satiety, **A**bdominal **P**ain, **O**bstuction, and in some cases acanthosis **N**igricans or Leser-Trélat sign (**WEAPON**).

- Intestinal—associated with *H pylori*, dietary nitrosamines (smoked foods common in East Asian countries), tobacco smoking, achlorhydria, chronic gastritis. Commonly on lesser curvature; looks like ulcer with raised margins.
- Diffuse—not associated with *H pylori*; most cases due to E-cadherin mutation; signet ring cells (mucin-filled cells with peripheral nuclei) **A**; stomach wall grossly thickened and leathery (linitis plastica).

Virchow node—involvement of left supraclavicular node by metastasis from stomach.

Krukenberg tumor—metastasis to ovaries (typically bilateral). Abundant mucin-secreting, signet ring cells.

Sister Mary Joseph nodule—subcutaneous periumbilical metastasis.

Blumer shelf—palpable mass on digital rectal exam suggesting metastasis to rectouterine pouch (pouch of Douglas).

Peptic ulcer disease

	Gastric ulcer	Duodenal ulcer
PAIN	Can be g reater with meals—weight loss	D ecreases with meals—weight gain
<i>H. PYLORI</i> INFECTION	~ 70%	~ 90%
MECHANISM	↓ mucosal protection against gastric acid	↓ mucosal protection or ↑ gastric acid secretion
OTHER CAUSES	NSAIDs	Zollinger-Ellison syndrome
RISK OF CARCINOMA	↑ Biopsy margins to rule out malignancy	Generally benign Not routinely biopsied

Ulcer complications**Hemorrhage**

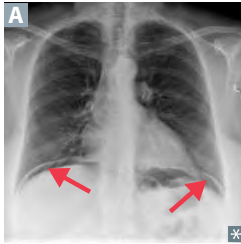
Gastric, duodenal (posterior > anterior). Most common complication.
 Ruptured gastric ulcer on the **l**esser curvature of stomach → bleeding from **l**eft gastric artery.
 An ulcer on the posterior wall of duodenum → bleeding from gastroduodenal artery.

Obstruction

Pyloric channel, duodenal.

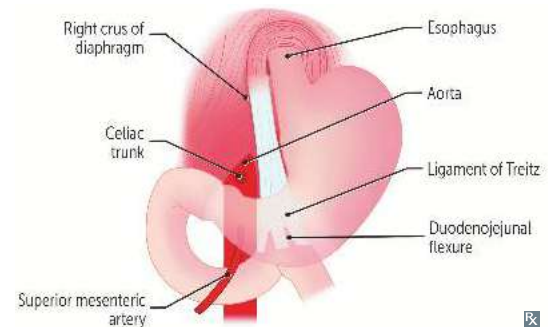
Perforation

Duodenal (anterior > posterior).
 Anterior duodenal ulcers can perforate into the anterior abdominal cavity, potentially leading to pneumoperitoneum.
 May see free air under diaphragm (pneumoperitoneum) **A** with referred pain to the shoulder via irritation of phrenic nerve.

**Acute gastrointestinal bleeding**

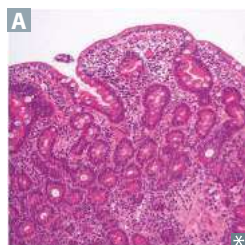
Upper GI bleeding—originates **proximal** to ligament of Treitz (suspensory ligament of duodenum). Usually presents with hematemesis and/or melena. Associated with peptic ulcer disease, variceal hemorrhage.

Lower GI bleeding—originates **distal** to ligament of Treitz. Usually presents with hematochezia. Associated with IBD, diverticulosis, angiodysplasia, hemorrhoids, anal fissure, cancer.



Malabsorption syndromes

Celiac disease



Can cause diarrhea, steatorrhea, weight loss, weakness, vitamin and mineral deficiencies. Screen for fecal fat (eg, Sudan stain).

Also called gluten-sensitive enteropathy, celiac sprue. Autoimmune-mediated intolerance of gliadin (gluten protein found in wheat, barley, rye). Associated with HLA-DQ2, HLA-DQ8 (I ate [8] too [2] much gluten at Dairy Queen), northern European descent.

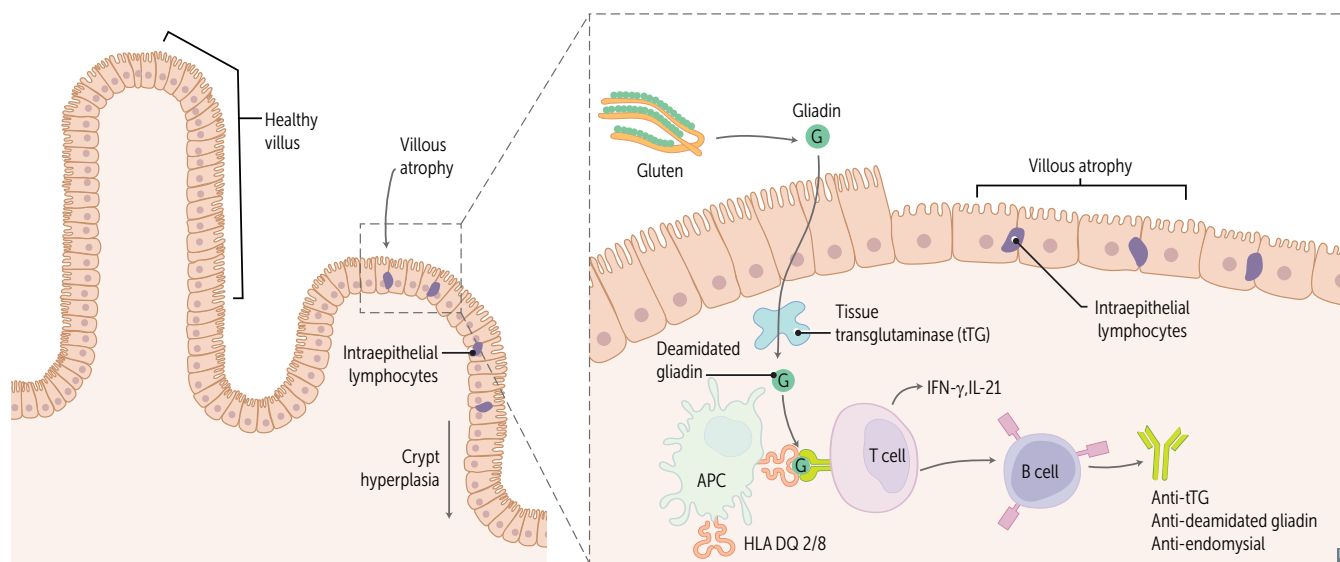
Primarily affects distal duodenum and/or proximal jejunum → malabsorption and steatorrhea.

Treatment: gluten-free diet.

Associated with dermatitis herpetiformis, ↓ bone density, iron deficiency anemia, moderately ↑ risk of malignancy (eg, T-cell lymphoma). D-xylose test: abnormal.

Serology: ⊕ IgA anti-tissue transglutaminase (IgA tTG), anti-endomysial, and anti-deamidated gliadin peptide antibodies.

Histology: Loss of villi, mucosal atrophy, crypt hyperplasia **A**, intraepithelial lymphocytosis.



Lactose intolerance

Lactase deficiency. Normal-appearing villi, except when 2° to injury at tips of villi (eg, viral enteritis). Osmotic diarrhea, ↓ stool pH (colonic bacteria ferment lactose).

Lactose hydrogen breath test: ⊕ for lactose malabsorption if post-lactose breath hydrogen value increases > 20 ppm compared with baseline.

Pancreatic insufficiency

Due to chronic pancreatitis, cystic fibrosis, obstructing cancer. Causes malabsorption of fat and fat-soluble vitamins (A, D, E, K) as well as vitamin B₁₂.

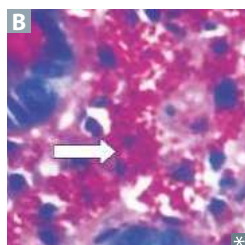
↓ duodenal bicarbonate (and pH) and fecal elastase.
D-xylose test: normal.

Tropical sprue

Similar findings as celiac sprue (affects small bowel), but responds to antibiotics. Cause is unknown, but seen in residents of or recent visitors to tropics.

↓ mucosal absorption affecting duodenum and jejunum but can involve ileum with time. Associated with megaloblastic anemia due to folate deficiency and, later, B₁₂ deficiency.

Whipple disease

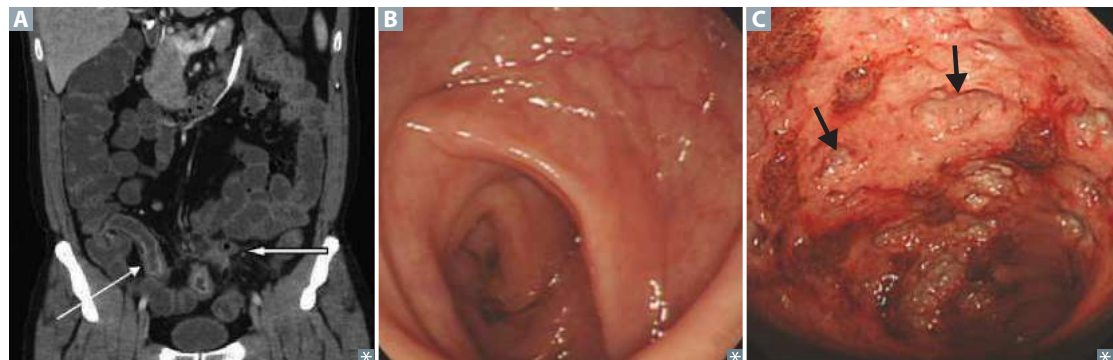


Infection with *Tropheryma whipplei* (intracellular gram ⊕); **PAS** ⊕ **foamy** macrophages in intestinal lamina propria **B**. Cardiac symptoms, Arthralgias, and Neurologic symptoms are common. Diarrhea/steatorrhea occur later in disease course. Most common in older males.

PASs the **foamy Whipped** cream in a **CAN**.

Inflammatory bowel diseases

	Crohn disease	Ulcerative colitis
LOCATION	Any portion of the GI tract, usually the terminal ileum and colon. Skip lesions, rectal sparing.	Colitis = colon inflammation. Continuous colonic lesions, always with rectal involvement.
GROSS MORPHOLOGY	Transmural inflammation → fistulas. Cobblestone mucosa, creeping fat, bowel wall thickening A (“string sign” on small bowel follow-through), linear ulcers, fissures.	Mucosal and submucosal inflammation only. Friable mucosa with superficial and/or deep ulcerations (compare normal B with diseased C). Loss of haustra → “lead pipe” appearance on imaging.
MICROSCOPIC MORPHOLOGY	Noncaseating granulomas, lymphoid aggregates.	Crypt abscesses/ulcers, bleeding, no granulomas.
COMPLICATIONS	Malabsorption/malnutrition, colorectal cancer (↑ risk with pancolitis). Fistulas (eg, enterovesical fistulae, which can cause recurrent UTI and pneumaturia), phlegmon/abscess, strictures (causing obstruction), perianal disease.	Fulminant colitis, toxic megacolon, perforation.
INTESTINAL MANIFESTATION	Diarrhea that may or may not be bloody.	Bloody diarrhea (usually painful).
EXTRAINTESTINAL MANIFESTATIONS	Rash (pyoderma gangrenosum, erythema nodosum), eye inflammation (episcleritis, uveitis), oral ulcerations (aphthous stomatitis), arthritis (peripheral, spondylitis). Calcium oxalate kidney stones—inflamed intestines → malabsorption of fat → fat binds calcium in intestine → excess free oxalate in intestines → ↑ oxalate reabsorption → combines with calcium in kidneys; gallstones. May be ⊕ for anti- <i>Saccharomyces cerevisiae</i> antibodies (ASCA).	1° sclerosing cholangitis. Associated with MPO-ANCA/p-ANCA.
TREATMENT	Glucocorticoids, azathioprine, antibiotics (eg, ciprofloxacin, metronidazole), biologics (eg, infliximab, adalimumab).	5-aminosalicylic acid preparations (eg, mesalamine), 6-mercaptopurine, infliximab, colectomy.
DISEASE ACTIVITY	Fecal calprotectin used to monitor activity and distinguish from noninflammatory diseases (irritable bowel).	

**Microscopic colitis**

Inflammatory disease of colon that causes chronic watery diarrhea. Most common in older females. Colonic mucosa appears normal on endoscopy. Histology shows lymphocytic infiltrate in lamina propria with intraepithelial lymphocytosis or thickened subepithelial collagen band.

Irritable bowel syndrome

Recurrent abdominal pain associated with ≥ 2 of the following:

- Related to defecation
- Change in stool frequency
- Change in form (consistency) of stool

No structural abnormalities. Most common in middle-aged females. Chronic symptoms may be diarrhea-predominant, constipation-predominant, or mixed. Pathophysiology is multifaceted. May be associated with fibromyalgia and mood disorders (anxiety, depression).

First-line treatment is lifestyle modification and dietary changes.

Appendicitis



Acute inflammation of the appendix (blue arrow in **A**), can be due to obstruction by fecalith (in adults) or lymphoid hyperplasia (in children).

Proximal appendiceal lumen obstruction $\rightarrow$ closed-loop obstruction $\rightarrow$ $\uparrow$ intraluminal pressure $\rightarrow$ stimulation of visceral afferent nerve fibers at T8-T10 $\rightarrow$ initial diffuse periumbilical pain $\rightarrow$ inflammation extends to serosa and irritates parietal peritoneum. Pain localized to RLQ/McBurney point (1/3 the distance from right anterior superior iliac spine to umbilicus). Nausea, fever; may perforate $\rightarrow$ peritonitis. May elicit psoas, obturator, and Rovsing (severe RLQ pain with palpation of LLQ) signs; guarding and rebound tenderness on exam.

Treatment: appendectomy.

Diverticula of the GI tract

Diverticulum

Blind pouch **A** protruding from the alimentary tract that communicates with the lumen of the gut. Most diverticula (esophagus, stomach, duodenum, colon) are acquired and are termed “false diverticula.”

“True” diverticulum—all gut wall layers outpouch (eg, Meckel).

“False” diverticulum or pseudodiverticulum—only mucosa and submucosa outpouch. Occur especially where vasa recta penetrate muscularis externa layer (eg, Zenker).

Diverticulosis

Many false diverticula of the colon **B**, commonly sigmoid. Common (in $\sim 50\%$ of people > 60 years). Caused by $\uparrow$ intraluminal pressure and focal weakness in colonic wall. Associated with obesity and diets low in fiber, high in total fat/red meat.

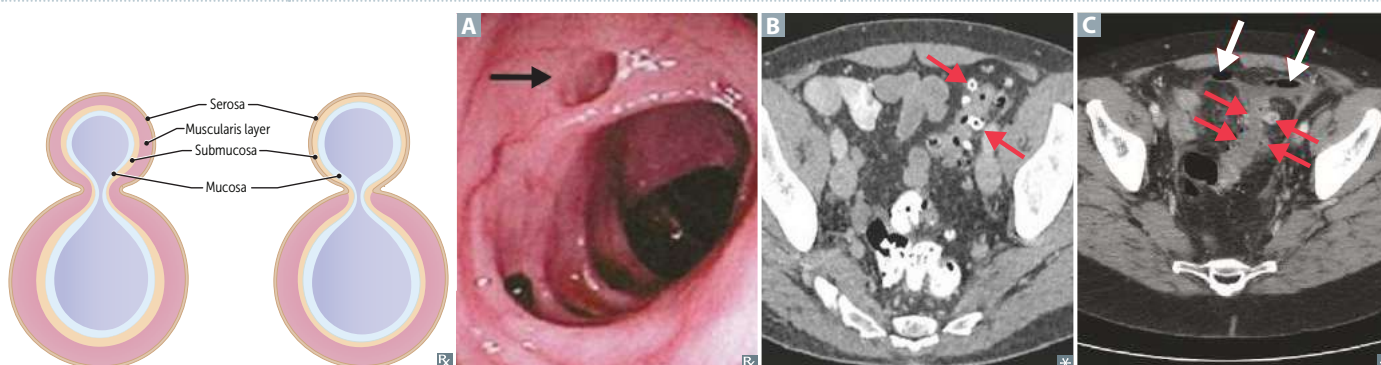
Often asymptomatic or associated with vague discomfort.

Complications include diverticular bleeding (painless hematochezia), diverticulitis.

Diverticulitis

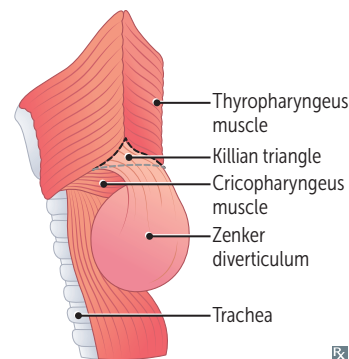
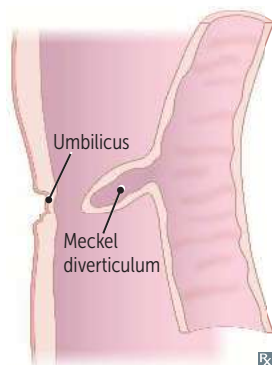
Inflammation of diverticula with wall thickening (red arrows in **C**) classically causing LLQ pain, fever, leukocytosis. Treat with supportive care (uncomplicated) or antibiotics (complicated).

Complications: abscess, fistula (colovesical fistula $\rightarrow$ pneumaturia), obstruction (inflammatory stenosis), perforation (white arrows in **C**) ($\rightarrow$ peritonitis). Hematochezia is rare.



Zenker diverticulum**Pharyngoesophageal false diverticulum A.**

Esophageal dysmotility causes herniation of mucosal tissue at an area of weakness between the thyropharyngeal and cricopharyngeal parts of the inferior pharyngeal constrictor (Killian triangle). Presenting symptoms: dysphagia, obstruction, gurgling, aspiration, foul breath, neck mass. Most common in older males.

**Meckel diverticulum**

True diverticulum. Persistence of the vitelline (omphalomesenteric) duct. May contain ectopic acid-secreting gastric mucosa and/or pancreatic tissue. Most common congenital anomaly of GI tract. Can cause painless hematochezia/melena (less common), RLQ pain, intussusception, volvulus, or obstruction near terminal ileum.

Diagnosis: ^{99m}Tc -pertechnetate scan (also called Meckel scan) for uptake by heterotopic gastric mucosa.

The rule of 2's:

2 times as likely in males.

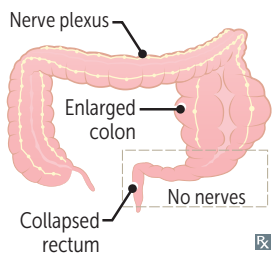
2 inches long.

2 feet from the ileocecal valve.

2% of population.

Commonly presents in first **2** years of life.

May have **2** types of epithelia (gastric/pancreatic).

Hirschsprung disease

Congenital megacolon characterized by lack of ganglion cells/enteric nervous plexuses (Auerbach and Meissner plexuses) in distal segment of colon. Due to failure of neural crest cell migration. Associated with loss of function mutations in *RET*.

Presents with bilious emesis, abdominal distention, and failure to pass meconium within 48 hours → chronic constipation.

Normal portion of the colon proximal to the aganglionic segment is dilated, resulting in a "transition zone."

Risk ↑ with Down syndrome.

Explosive expulsion of feces (squirt sign)
→ empty rectum on digital exam.

Diagnosed by absence of ganglion cells on rectal suction biopsy.

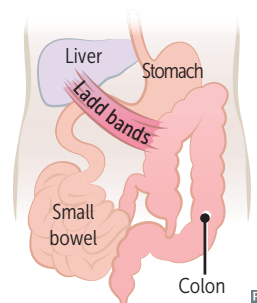
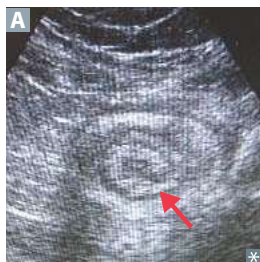
Treatment: resection.

RET mutation in the **REcTum**.

Malrotation

Anomaly of midgut rotation during fetal development → improper positioning of bowel (small bowel clumped on the right side and colon on the left), formation of fibrous bands (Ladd bands).

Can lead to volvulus, duodenal obstruction.

**Intussusception**

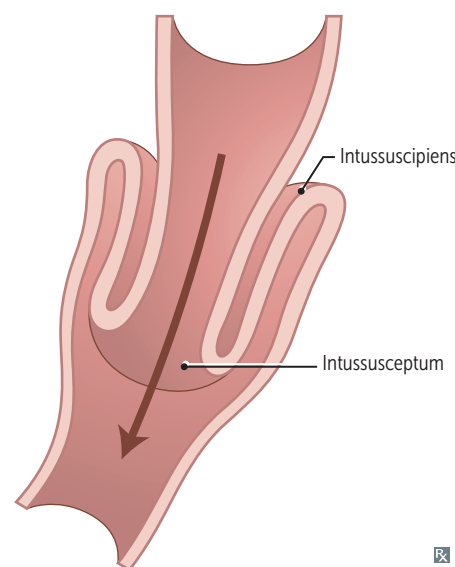
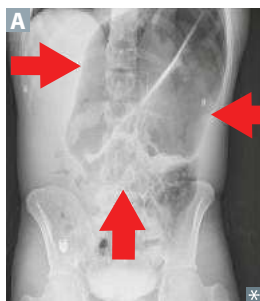
Telescoping of a proximal bowel segment into a distal segment, most commonly at ileocecal junction. Typically seen in infants.

Usually idiopathic in children, less frequently due to an identifiable lead point. Idiopathic form is associated with recent viral infections (eg, adenovirus), rotavirus vaccine → Peyer patch hypertrophy may act as a lead point. Common lead points:

- Children—Meckel diverticulum, small bowel wall hematoma (IgA vasculitis).
- Adults—intraluminal mass/tumor.

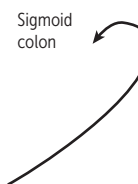
Causes small bowel obstruction and vascular compromise → intermittent abdominal pain, vomiting, bloody “currant jelly” stools. Sausage-shaped mass in right abdomen on exam. Patient may draw their legs to chest to ease pain.

Ultrasound/CT may show “target sign” **A**.

**Volvulus**

Twisting of portion of bowel around its mesentery; can lead to obstruction and infarction. Can occur throughout the GI tract.

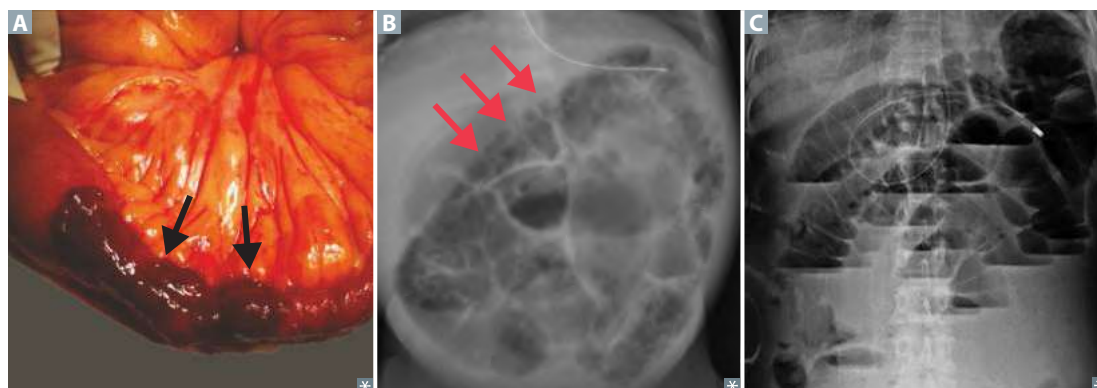
- Gastric volvulus more common with abnormalities (paraesophageal hernia) in adults, and presents with severe abdominal pain, dry heaving, and inability to pass nasogastric tube
- Midgut volvulus more common in infants and children (minors)
- Sigmoid volvulus (coffee bean sign on x-ray **A**) more common in older adults (seniors)

**Short bowel syndrome**

Inability to adequately absorb nutrients in the small intestine 2° to significant surgical resection (eg, Crohn disease, malignancy, trauma). Malabsorption of bile salts and fat at the distal ileum → postprandial voluminous diarrhea, dehydration, weight loss, anemia, calcium oxalate kidney stones.

Other intestinal disorders

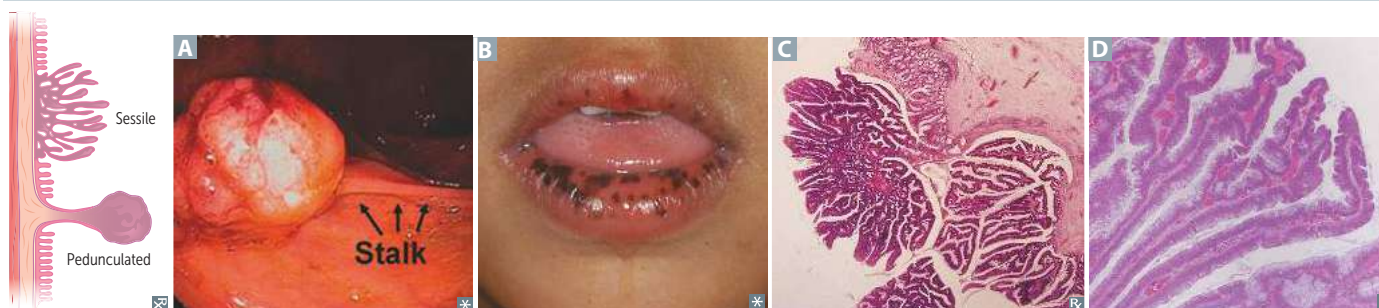
Acute mesenteric ischemia	Critical blockage of intestinal blood flow (often embolic occlusion of SMA) → small bowel necrosis A → abdominal pain out of proportion to physical findings. May see red “currant jelly” stools. Risk factors: atrial fibrillation, peripheral arterial disease, recent MI, CHF.
Angiodysplasia	Tortuous dilation of vessels → hematochezia. Most often found in the right-sided colon. More common in older patients. Confirmed by angiography. Associated with end-stage renal disease, von Willebrand disease, aortic stenosis.
Chronic mesenteric ischemia	“Intestinal angina”: atherosclerosis of celiac artery, SMA (most commonly affected), or IMA → intestinal hypoperfusion → postprandial epigastric pain → food aversion and weight loss.
Colonic ischemia	Crampy abdominal pain followed by hematochezia. Commonly occurs at watershed areas (splenic flexure, rectosigmoid junction). Typically affects older adults. Thumbprint sign on imaging due to mucosal edema/hemorrhage.
Ileus	Intestinal hypomotility without obstruction → constipation and ↓ flatus; distended/tympanic abdomen with ↓ bowel sounds. Associated with abdominal surgeries, opiates, hypokalemia, sepsis. No transition zone on imaging. Treatment: bowel rest, electrolyte correction, cholinergic drugs (stimulate intestinal motility).
Necrotizing enterocolitis	Seen in premature, formula-fed infants with immature immune system. Necrosis of intestinal mucosa (most commonly terminal ileum and proximal colon), which can lead to pneumatosis intestinalis (arrows in B), pneumoperitoneum, portal venous gas.
Proctitis	Inflammation of rectal mucosa, usually associated with infection (<i>N gonorrhea</i> , <i>Chlamydia</i> , <i>Campylobacter</i> , <i>Shigella</i> , <i>Salmonella</i> , HSV, CMV), IBD, or radiation. Patients report tenesmus, rectal bleeding, and rectal pain. Proctoscopy reveals inflamed rectal mucosa (ulcers/vesicles in the case of HSV). Rectal swabs are used to detect other infectious etiologies.
Small bowel obstruction	Normal flow of intraluminal contents is interrupted → fluid accumulation and intestinal dilation proximal to blockage and intestinal decompression distal to blockage. Presents with abrupt onset of abdominal pain, nausea, vomiting, abdominal distension. Compromised blood flow due to excessive dilation or strangulation may lead to ischemia, necrosis, or perforation. Most commonly caused by intraperitoneal adhesions (fibrous band of scar tissue), tumors, and hernias (in rare cases, meconium plug in newborns → meconium ileus). Upright abdominal x-ray shows air-fluid levels C . Management: gastrointestinal decompression, volume resuscitation, bowel rest.
Small intestinal bacterial overgrowth	Abnormal bacterial overgrowth in the small intestine (normally low bacterial colony count). Risk factors: altered pH (eg, achlorhydria, PPI use), anatomical (eg, small bowel obstruction, adhesions, fistula, gastric bypass surgery, blind loop), dysmotility (eg, gastroparesis), immune mediated (IgA deficiency, HIV). Presents with bloating, flatulence, abdominal pain, chronic watery diarrhea, malabsorption (vitamin B ₁₂) in severe cases. Diagnosis: carbohydrate breath test or small bowel culture.



Colonic polyps

Growths of tissue within the colon **A**. Grossly characterized as flat, sessile, or pedunculated on the basis of protrusion into colonic lumen. Generally classified by histologic type.

HISTOLOGIC TYPE	CHARACTERISTICS
Generally nonneoplastic	
Hamartomatous polyps	Solitary lesions do not have significant risk of transformation. Growths of normal colonic tissue with distorted architecture. Associated with Peutz-Jeghers syndrome B and juvenile polyposis.
Hyperplastic polyps	Most common; generally smaller and predominantly located in rectosigmoid region. Occasionally evolve into serrated polyps and more advanced lesions.
Inflammatory pseudopolyps	Due to mucosal erosion in inflammatory bowel disease.
Mucosal polyps	Small, usually < 5 mm. Look similar to normal mucosa. Clinically insignificant.
Submucosal polyps	May include lipomas, leiomyomas, fibromas, and other lesions.
Potentially malignant	
Adenomatous polyps	Neoplastic, via chromosomal instability pathway with mutations in <i>APC</i> and <i>KRAS</i> . Tubular C histology has less malignant potential than villous D (“ villous ” histology is villainous ”); tubulovillous has intermediate malignant potential. Usually asymptomatic; may present with occult bleeding.
Serrated polyps	Neoplastic. Characterized by CpG island methylator phenotype (CIMP; cytosine base followed by guanine, linked by a phosphodiester bond). Defect may silence mismatch repair gene (eg, <i>MLH1</i>) expression. Mutations lead to microsatellite instability and mutations in <i>BRAF</i> . “Saw-tooth” pattern of crypts on biopsy. Up to 20% of cases of sporadic CRC.

**Polyposis syndromes**

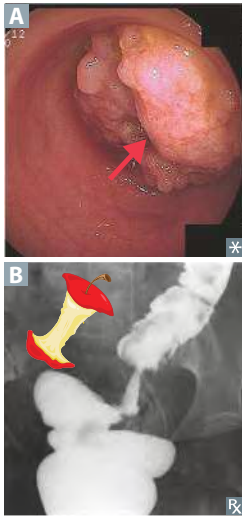
Familial adenomatous polyposis	Autosomal dominant mutation of <i>APC</i> tumor suppressor gene on chromosome 5q21-q22. 2-hit hypothesis. Thousands of polyps arise starting after puberty; pancolonic; always involves rectum. Prophylactic colectomy or else 100% progress to CRC.
Gardner syndrome	FAP + osseous and soft tissue tumors (eg, osteomas of skull or mandible), congenital hypertrophy of retinal pigment epithelium, impacted/supernumerary teeth.
Turcot syndrome	FAP or Lynch syndrome + malignant CNS tumor (eg, medulloblastoma, glioma). Turcot = Turban .
Peutz-Jeghers syndrome	Autosomal dominant syndrome featuring numerous hamartomatous polyps throughout GI tract, along with hyperpigmented macules on mouth, lips, hands, genitalia. Associated with ↑ risk of breast and GI cancers (eg, colorectal, stomach, small bowel, pancreatic).
Juvenile polyposis syndrome	Autosomal dominant syndrome in children (typically < 5 years old) featuring numerous hamartomatous polyps in the colon, stomach, small bowel. Associated with ↑ risk of CRC.
MUTYH-associated polyposis syndrome	Autosomal recessive disorder of the <i>MUTYH</i> gene responsible for DNA repair. Associated with significantly ↑ risk of CRC, polyps (adenomatous; may be hyperplastic or serrated), and serrated adenomas. Also associated with duodenal adenomas, ovarian and bladder cancers.

Lynch syndrome

Also called hereditary nonpolyposis colorectal cancer (HNPCC). Autosomal dominant mutation of mismatch repair genes (eg, *MLH1*, *MSH2*) with subsequent microsatellite instability. ~80% progress to CRC. Proximal Colon is always involved. Associated with Endometrial, Ovarian, and Skin cancers. Merrill Lynch has CEOS.

Colorectal cancer

EPIDEMIOLOGY	Most patients are > 50 years old. ~25% have a family history.
RISK FACTORS	Adenomatous and serrated polyps, familial cancer syndromes, IBD, tobacco use, diet of processed meat with low fiber.
PRESENTATION	Rectosigmoid > ascending > descending. Most are asymptomatic. Right side (cecal, ascending) associated with occult bleeding; left side (rectosigmoid) associated with hematochezia and obstruction (narrower lumen → ↓ stool caliber). Ascending—exophytic mass, iron deficiency anemia, weight loss. Descending—infiltrating mass, partial obstruction, colicky pain, hematochezia. Can present with <i>S bovis</i> (<i>gallolyticus</i>) bacteremia/endocarditis or as an episode of diverticulitis.
DIAGNOSIS	Iron deficiency anemia in males (especially > 50 years old) and postmenopausal females raises suspicion. Screening: - Average risk: screen at age 45 with colonoscopy (polyp seen in A); alternatives include flexible sigmoidoscopy, fecal occult blood testing (FOBT), fecal immunochemical testing (FIT), FIT-fecal DNA, CT colonography. - Patients with a first-degree relative who has colon cancer: screen at age 40 with colonoscopy, or 10 years prior to the relative's presentation. - Patients with IBD: screen 8 years after onset. “Apple core” lesion seen on barium enema x-ray B. CEA tumor marker: good for monitoring recurrence, should not be used for screening.

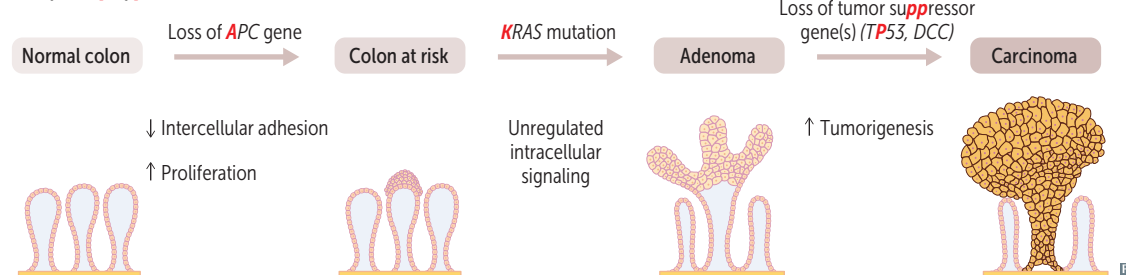
**Molecular pathogenesis of colorectal cancer**

Chromosomal instability pathway: mutations in *APC* cause FAP and most sporadic cases of CRC (commonly left-sided) via adenoma-carcinoma sequence.

Microsatellite instability pathway: mutations or methylation of mismatch repair genes (eg, *MLH1*) cause Lynch syndrome and some sporadic CRC (commonly right sided) via serrated polyp pathway. Overexpression of COX-2 has been linked to CRC, NSAIDs may be chemopreventive.

Chromosomal instability pathway

Always kill polyps



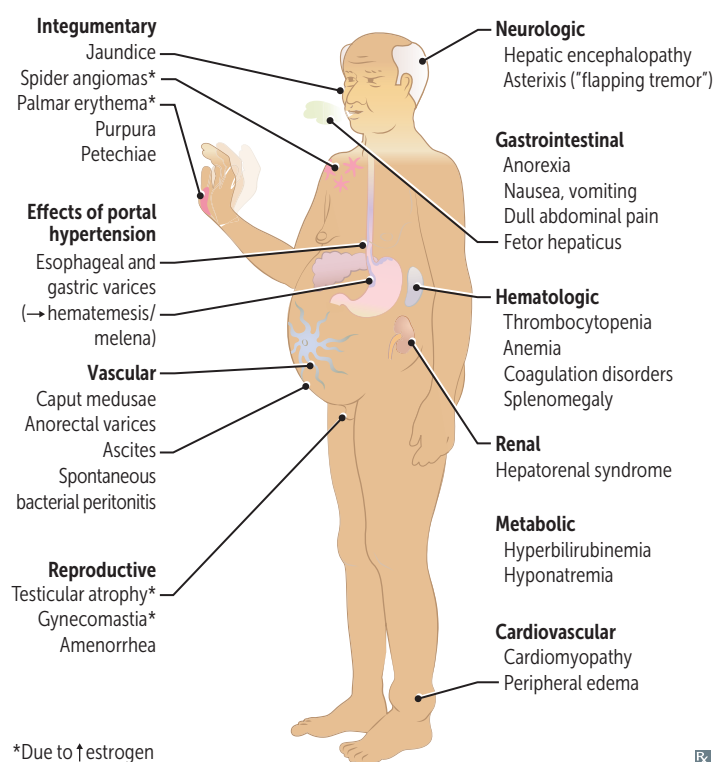
Cirrhosis and portal hypertension



Cirrhosis—diffuse bridging fibrosis (via stellate cells) and regenerative nodules disrupt normal architecture of liver **A**; ↑ risk for hepatocellular carcinoma. Can lead to various systemic changes. Etiologies include alcohol, nonalcoholic steatohepatitis, chronic viral hepatitis, autoimmune hepatitis, biliary disease, genetic/metabolic disorders.

Portal hypertension—↑ pressure in portal vein system → new collateral circulation → collateral vessels → varices (ie, esophageal, gastric, and rectal). Causes: cirrhosis (most common), schistosomiasis, portal vein thrombosis.

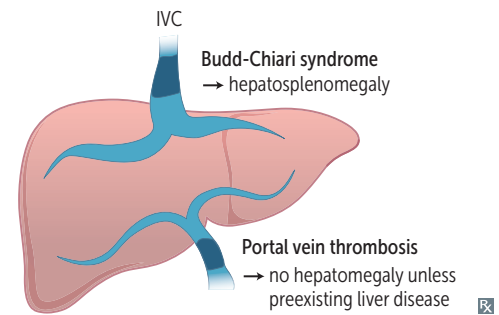
Ascites—pathological fluid accumulation in the peritoneal cavity due to portal hypertension or portal vein thrombosis. ↑ hepatic sinusoidal pressure → ascites. Serum-to-ascites albumin gradient (SAAG) analysis of ascitic fluid sample determines between etiologies of ascites. SAAG ≥ 1.1 = portal hypertension.



Budd-Chiari syndrome

Hepatic venous outflow tract obstruction (eg, due to thrombosis, compression) with centrilobular congestion and necrosis → congestive liver disease (hepatomegaly, ascites, varices, abdominal pain, liver failure). Absence of JVD. Associated with hypercoagulable states, polycythemia vera, postpartum state, HCC. May cause nutmeg liver (mottled appearance).

Portal vein thrombosis—thrombosis in portal vein proximal to liver. Usually asymptomatic in the majority of patients, but associated with portal hypertension, abdominal pain, fever. May lead to bowel ischemia if extension to superior mesenteric vein. Etiologies include cirrhosis, malignancy, pancreatitis, and sepsis.

**Spontaneous bacterial peritonitis**

Also called 1° bacterial peritonitis. Common and potentially fatal bacterial infection in patients with cirrhosis and ascites. Often asymptomatic, but can cause fevers, chills, abdominal pain, ileus, or worsening encephalopathy. Commonly caused by gram $\ominus$ organisms (eg, *E coli*, *Klebsiella*) or less commonly gram $\oplus$ *Streptococcus*.

Diagnosis: paracentesis with ascitic fluid absolute neutrophil count (ANC) > 250 cells/mm³. Empiric first-line treatment is 3rd generation cephalosporin (eg, ceftriaxone).

Serum markers of liver pathology

ENZYMES RELEASED IN LIVER DAMAGE

Aspartate aminotransferase and alanine aminotransferase

↑ in most liver disease: ALT > AST

↑ in **alcoholic** liver disease: **AST** > ALT (ratio usually > 2:1, AST does not typically exceed 500 U/L in alcoholic hepatitis). Make a to**AST** with **alcohol**

AST > ALT in nonalcoholic liver disease suggests progression to advanced fibrosis or cirrhosis

↑↑↑ aminotransferases (>1000 U/L): differential includes drug-induced liver injury (eg, acetaminophen toxicity), ischemic hepatitis, acute viral hepatitis, autoimmune hepatitis

Alkaline phosphatase

↑ in cholestasis (eg, biliary obstruction), infiltrative disorders, bone disease

γ-glutamyl transpeptidase

↑ in various liver and biliary diseases (just as ALP can), but not in bone disease (located in canalicular membrane of hepatocytes like ALP); associated with alcohol use

FUNCTIONAL LIVER MARKERS

Bilirubin

↑ in various liver diseases (eg, biliary obstruction, alcoholic or viral hepatitis, cirrhosis), hemolysis

Albumin

↓ in advanced liver disease (marker of liver's biosynthetic function)

Prothrombin time

↑ in advanced liver disease (↓ production of clotting factors, thereby measuring the liver's biosynthetic function)

Platelets

↓ in advanced liver disease (↓ thrombopoietin, liver sequestration) and portal hypertension (splenomegaly/splenic sequestration)

Reye syndrome

Rare, often fatal childhood hepatic encephalopathy.

Associated with viral infection (especially VZV and influenza) that has been treated with aspirin. Aspirin metabolites ↓ β -oxidation by reversible inhibition of mitochondrial enzymes.

Findings: mitochondrial abnormalities, fatty liver (microvesicular fatty changes), hyperammonemia, hypoglycemia, vomiting, hepatomegaly, coma.

↑ ICP ↑ morbidity and mortality. Renal and cardiac failure may also occur.

Avoid aspirin (**ASA**) in children, except in Kaw**ASA**ki disease.

Salicylates aren't a ray (**Reye**) of sun**SHINEE** for kids:

Steatosis of liver/hepatocytes

Hypoglycemia/**H**epatomegaly

Infection (VZV, influenza)

Not awake (coma)

Encephalopathy and diffuse cerebral **E**dema

Alcoholic liver disease**Alcoholic liver disease**

Excess NADH production → ↓ fatty acid oxidation and ↑ lipogenesis.

Hepatic steatosis

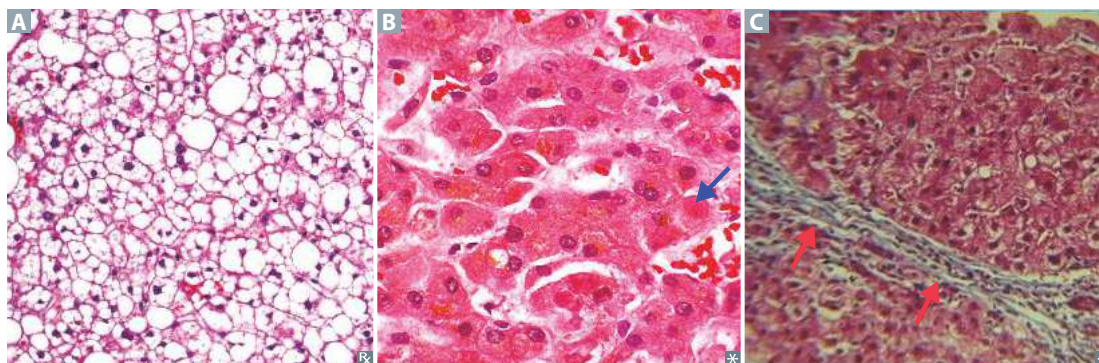
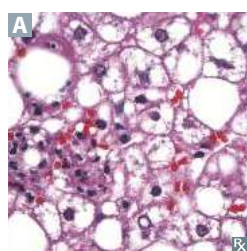
Macrovesicular fatty change **A**; may be reversible with alcohol cessation.

Alcoholic hepatitis

Requires sustained, long-term consumption. Swollen and necrotic hepatocytes with neutrophilic infiltration. Mallory bodies **B** (intracytoplasmic eosinophilic inclusions of damaged keratin filaments).

Alcoholic cirrhosis

Final and usually irreversible form. Sclerosis around central vein may be seen in early disease. Regenerative nodules surrounded by fibrous bands (red arrows in **C**) in response to chronic liver injury → portal hypertension and end-stage liver disease.

**Steatotic liver disease**

Steatotic liver disease (SLD) encompasses metabolic dysfunction–associated SLD (MASLD; formerly known as nonalcoholic fatty liver disease), MASLD and increased alcohol intake, alcohol-associated liver disease, specific etiology SLD, and cryptogenic SLD.

MASLD is associated with metabolic syndrome (obesity, insulin resistance, HTN, hypertriglyceridemia, ↓ HDL); obesity → fatty infiltration of hepatocytes **A** → cellular “ballooning” and eventual necrosis. Steatosis present without evidence of significant inflammation or fibrosis. May persist or even regress over time. Usually asymptomatic.

Metabolic dysfunction–associated steatohepatitis—associated with lobular inflammation and hepatocyte ballooning → fibrosis. May progress to cirrhosis and HCC.

Autoimmune hepatitis Chronic inflammatory liver disease. More common in females. May be asymptomatic or present with fatigue, nausea, pruritus. Often $\oplus$ for anti-smooth muscle or anti-liver/kidney microsomal-1 antibodies. Labs: $\uparrow$ ALT and AST. Histology: portal and periportal lymphoplasmacytic infiltrate.

Hepatic encephalopathy Cirrhosis $\rightarrow$ portosystemic shunts $\rightarrow$ $\downarrow$ NH_3 metabolism $\rightarrow$ neuropsychiatric dysfunction (reversible) ranging from disorientation/asterixis to difficult arousal or coma.
Triggers:

- $\uparrow$ NH_3 production and absorption (due to GI bleed, constipation, infection).
- $\downarrow$ NH_3 removal (due to renal failure, diuretics, bypassed hepatic blood flow post-TIPS).

 Treatment: lactulose ($\uparrow$ NH_4^+ generation) and rifaximin ($\downarrow$ NH_3 -producing gut bacteria).

Liver tumors

Hepatic hemangioma Also called cavernous hemangioma. Most common benign liver tumor (venous malformation) **A**; typically occurs at age 30–50 years. Biopsy contraindicated because of risk of hemorrhage.

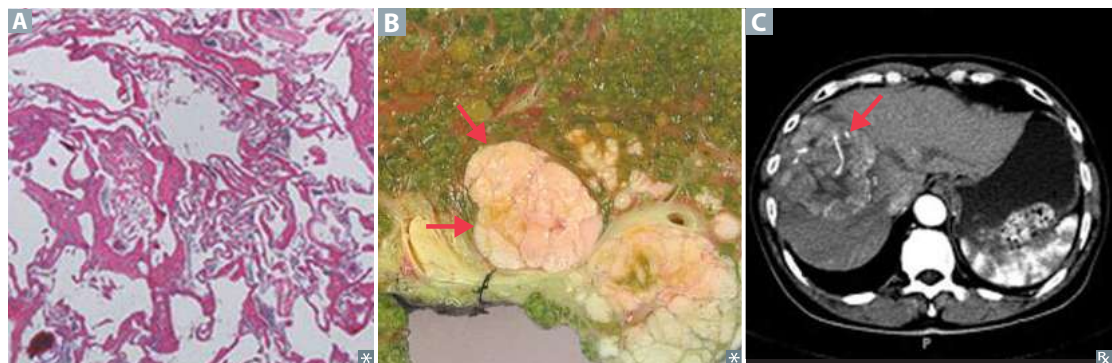
Focal nodular hyperplasia Second most common benign liver tumor; occurs predominantly in females aged 35–50 years. Hyperplastic reaction of hepatocytes to an aberrant dystrophic artery. Marked by central stellate scar. Usually asymptomatic and detected incidentally.

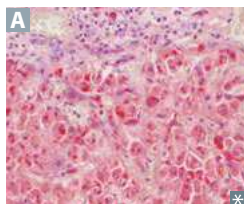
Hepatic adenoma Rare, benign tumor, often related to oral contraceptive or anabolic steroid use; may regress spontaneously or rupture (abdominal pain and shock).

Hepatocellular carcinoma Also called hepatoma. Most common 1° malignant liver tumor in adults **B**. Associated with HBV ($\pm$ cirrhosis) and all other causes of cirrhosis (including HCV, alcoholic and nonalcoholic fatty liver disease, autoimmune disease, hemochromatosis, Wilson disease, α_1 -antitrypsin deficiency) and specific carcinogens (eg, aflatoxin from *Aspergillus*).
Findings: anorexia, jaundice, tender hepatomegaly. May lead to decompensation of previously stable cirrhosis (eg, ascites) and portal vein thrombosis. Spreads hematogenously.
Diagnosis: ultrasound (screening) or contrast CT/MRI **C** (confirmation); biopsy if diagnosis is uncertain. Recurrence monitored with serum AFP.

Hepatic angiosarcoma Rare, malignant tumor of endothelial origin; associated with exposure to arsenic, vinyl chloride.

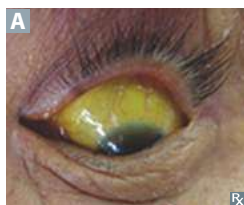
Metastases Most common malignant liver tumors overall; 1° sources include GI, breast, lung cancers. Metastases are rarely solitary.



α_1 -antitrypsin deficiency

Misfolded gene product protein aggregates in hepatocellular ER → cirrhosis with PAS ⊕ globules **A** in liver. Codominant trait. Often presents in young patients with liver damage and dyspnea without a history of tobacco smoking.

In lungs, ↓ α_1 -antitrypsin → uninhibited elastase in alveoli → ↓ elastic tissue → panacinar emphysema.

Jaundice

Abnormal yellowing of the skin and/or sclera (icterus) **A** due to bilirubin deposition. Hyperbilirubinemia 2° to ↑ production or ↓ clearance (impaired hepatic uptake, conjugation, excretion).

HOT Liver—common causes of ↑ bilirubin level:

Hemolysis
Obstuction
Tumor
Liver disease

Conjugated (direct) hyperbilirubinemia

Biliary tract obstruction: gallstones, cholangiocarcinoma, pancreatic or liver cancer, liver fluke.
Biliary tract disease: 1° sclerosing cholangitis, 1° biliary cholangitis
Excretion defect: Dubin-Johnson syndrome, Rotor syndrome.

Unconjugated (indirect) hyperbilirubinemia

Hemolytic, benign (neonates), Crigler-Najjar, Gilbert syndrome.

Mixed hyperbilirubinemia

Both direct and indirect hyperbilirubinemia.
Hepatitis, cirrhosis.

Benign neonatal hyperbilirubinemia

Formerly called physiologic neonatal jaundice. Mild unconjugated hyperbilirubinemia caused by:

- ↑ fetal RBC turnover (↑ hematocrit and ↓ fetal RBC lifespan).
- Immature newborn liver (↓ UDP-glucuronosyltransferase activity).
- Sterile newborn gut (↓ conversion to urobilinogen → ↑ deconjugation by intestinal brush border β -glucuronidase → ↑ enterohepatic circulation).

β -glucuronidase—lysosomal enzyme for direct bilirubin deconjugation. Also found in breast milk. May lead to pigment stone formation.

Occurs in nearly all newborns after first 24 hours of life and usually resolves without treatment in 1–2 weeks. Exaggerated forms:

Breastfeeding failure jaundice—insufficient breast milk intake → ↓ bilirubin elimination in stool → ↑ enterohepatic circulation.

Breast milk jaundice—↑ β -glucuronidase in breast milk → ↑ deconjugation → ↑ enterohepatic circulation.

Severe cases may lead to kernicterus (deposition of unconjugated, lipid-soluble bilirubin in the brain, particularly basal ganglia).

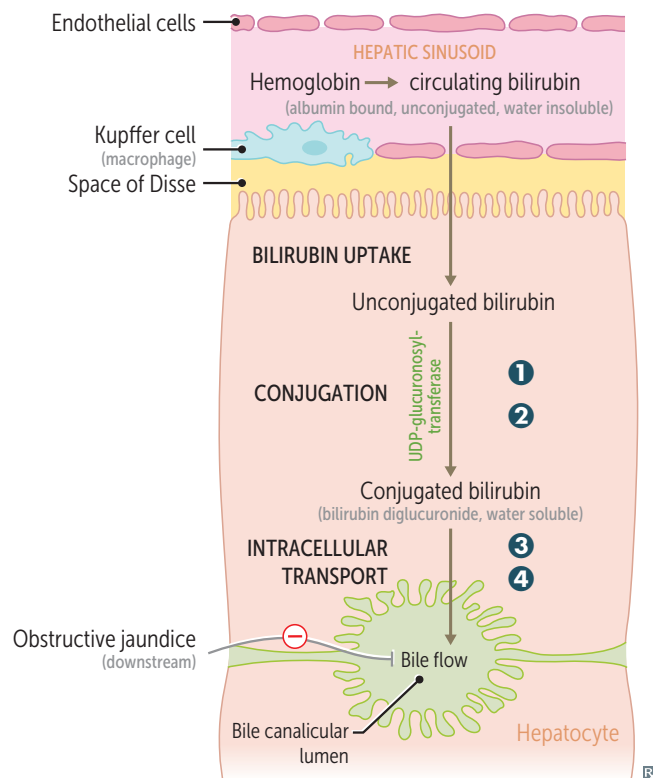
Treatment: phototherapy (non-UV) isomerizes unconjugated bilirubin to water-soluble form that can be excreted in the bile.

Biliary atresia

Most common reason for pediatric liver transplantation. Fibro-obliterative destruction of bile ducts → cholestasis. Associated with absent/abnormal gallbladder on ultrasonogram. Often presents as a newborn with persistent jaundice after 2 weeks of life, darkening urine, acholic stools, hepatomegaly. Labs: ↑ direct bilirubin and GGT.

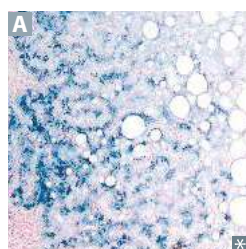
Hereditary hyperbilirubinemias

- | | |
|--|--|
| ① Gilbert syndrome | Mildly ↓ UDP-glucuronosyltransferase conjugation. Asymptomatic or mild jaundice usually with stress, illness, or fasting. ↑ unconjugated bilirubin without overt hemolysis. Relatively common, benign condition. |
| ② Crigler-Najjar syndrome, type I | Absent UDP-glucuronosyltransferase. Presents early in life, but some patients may not have neurologic signs until later in life. Findings: jaundice, kernicterus (unconjugated bilirubin deposition in brain), ↑ unconjugated bilirubin. Treatment: plasmapheresis and phototherapy (does not conjugate UCB; but does ↑ polarity and ↑ water solubility to allow excretion). Liver transplant is curative. Type II is less severe and responds to phenobarbital (vs. Type I, more severe), which ↑ liver enzyme synthesis. |
| ③ Dubin-Johnson syndrome | Conjugated hyperbilirubinemia due to defective liver excretion. Grossly black (D ark) liver due to impaired excretion of epinephrine metabolites. Benign. |
| ④ Rotor syndrome | Phenotypically similar to Dubin-Johnson, but milder in presentation without black (R egular) liver. Due to impaired hepatic storage of conjugated bilirubin. |



Wilson disease

Also called hepatolenticular degeneration. Autosomal recessive mutations in hepatocyte copper-transporting ATPase (*ATP7B* gene; chromosome 13) → ↓ copper incorporation into apoceruloplasmin and excretion into bile → ↓ serum ceruloplasmin. Copper accumulates, especially in liver, brain (eg, basal ganglia), cornea, kidneys; ↑ urine copper. Presents before age 40 with liver disease (eg, hepatitis, acute liver failure, cirrhosis), neurologic disease (eg, dysarthria, dystonia, tremor, parkinsonism), psychiatric disease, Kayser-Fleischer rings (deposits in Descemet membrane of cornea) **A**, hemolytic anemia, renal disease (eg, Fanconi syndrome). Treatment: chelation with penicillamine or trientine, oral zinc. Liver transplant in acute liver failure related to Wilson disease.

Hemochromatosis

Autosomal recessive. Mutation in *HFE* gene, located on chromosome 6. Leads to abnormal (low) hepcidin production, ↑ intestinal iron absorption. Iron overload can also be 2° to chronic transfusion therapy (eg, β-thalassemia major). Iron accumulates, especially in liver, pancreas, skin, heart, pituitary, joints. Hemosiderin (iron) can be identified on liver MRI or biopsy with Prussian blue stain **A**. Presents after age 40 when total body iron > 20 g; iron loss through menstruation slows progression in females. Classic triad of cirrhosis, diabetes mellitus, skin pigmentation (“bronze diabetes”). Also causes restrictive cardiomyopathy (classic) or dilated cardiomyopathy (reversible), hypogonadism, arthropathy (calcium pyrophosphate deposition; especially metacarpophalangeal joints). HCC is common cause of death. Treatment: repeated phlebotomy, iron (**Fe**) chelation with deferasirox, deferoxamine, deferiprone.

Biliary tract disease

May present with pruritus, jaundice, dark urine, light-colored stool, hepatosplenomegaly. Typically with cholestatic pattern of LFTs (↑ conjugated bilirubin, ↑ cholesterol, ↑ ALP, ↑ GGT).

	PATHOLOGY	EPIDEMIOLOGY	ADDITIONAL FEATURES
Primary sclerosing cholangitis 	Unknown cause of concentric “onion skin” bile duct fibrosis → alternating strictures and dilation with “beading” of intra- and extrahepatic bile ducts on ERCP A , magnetic resonance cholangiopancreatography (MRCP).	Classically in middle-aged males with ulcerative colitis.	Associated with ulcerative colitis. MPO-ANCA/ p-ANCA ⊕. ↑ IgM. Can lead to 2° biliary cirrhosis. ↑ risk of cholangiocarcinoma and gallbladder cancer.
Primary biliary cholangitis	Autoimmune reaction → lymphocytic infiltrate +/- granulomas → destruction of lobular bile ducts.	Classically in middle-aged females.	Antimitochondrial antibody ⊕, ↑ IgM. Associated with other autoimmune conditions (eg, Hashimoto thyroiditis, rheumatoid arthritis, celiac disease). Treatment: ursodiol.
Secondary biliary cirrhosis	Extrahepatic biliary obstruction → ↑ pressure in intrahepatic ducts → injury/ fibrosis and bile stasis.	Patients with known obstructive lesions (gallstones, biliary strictures, pancreatic carcinoma).	May be complicated by acute cholangitis.

Cholelithiasis and related pathologies



↑ cholesterol and/or bilirubin, ↓ bile salts, and gallbladder stasis all cause sludge or stones.

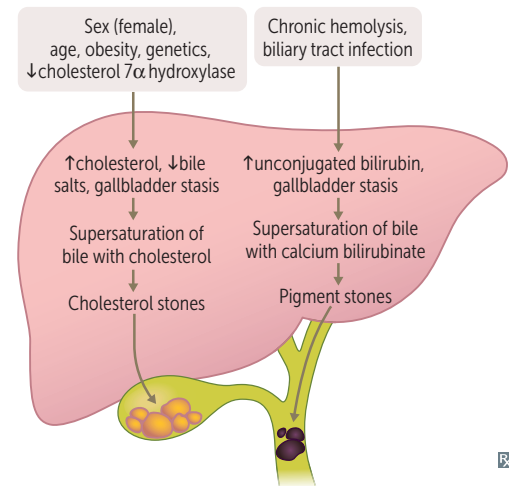
2 types of stones:

- Cholesterol stones **A** (radiolucent with 10–20% opaque due to calcifications)—80% of stones. Associated with obesity, Crohn disease, advanced age, estrogen therapy, multiparity, rapid weight loss, medications (eg, fibrates), race (↑ incidence in White and Native American populations).
- Pigment stones (black = radiopaque, Ca^{2+} bilirubinate, hemolysis; brown = radiolucent, infection). Associated with Crohn disease, chronic hemolysis, alcoholic cirrhosis, advanced age, biliary infections, total parenteral nutrition (TPN).

Most common complication is cholecystitis; can also cause acute pancreatitis, acute cholangitis.

Diagnose with ultrasound. Treat with elective cholecystectomy if symptomatic.

Risk factors (**8 F's**): female, fat, fertile, forty, fair, feeds (TPN), fasting (rapid weight loss), fibrates.



RELATED PATHOLOGIES

CHARACTERISTICS

Biliary colic

Associated with nausea/vomiting and dull RUQ pain. Neurohormonal activation (eg, by CCK after a fatty meal) triggers contraction of gallbladder, forcing stone into cystic duct. Labs are normal, ultrasound shows cholelithiasis.

Choledocholithiasis

Presence of gallstone(s) in common bile duct, often leading to elevated ALP, GGT, direct bilirubin, and/or AST/ALT.

Cholecystitis



Acute or chronic inflammation of gallbladder.

Calculous cholecystitis—most common type; due to gallstone impaction in the cystic duct resulting in inflammation and gallbladder wall thickening (arrows in **B**); can produce 2° infection.

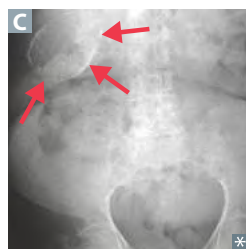
Acalculous cholecystitis—due to gallbladder stasis, hypoperfusion, or infection (CMV); seen in critically ill patients.

Murphy sign: inspiratory arrest on RUQ palpation due to pain. Pain may radiate to right shoulder (due to irritation of phrenic nerve). ↑ ALP if bile duct becomes involved (eg, acute cholangitis).

Diagnose with ultrasound or cholescintigraphy (HIDA scan). Failure to visualize gallbladder on HIDA scan suggests obstruction.

Gallstone ileus—fistula between gallbladder and GI tract → stone enters GI lumen → obstructs at ileocecal valve (narrowest point); can see air in biliary tree (pneumobilia). Rigler triad: radiographic findings of pneumobilia, small bowel obstruction, gallstone (usually in iliac fossa).

Porcelain gallbladder



Calcified gallbladder due to chronic cholecystitis; usually found incidentally on imaging **C**.

Treatment: prophylactic cholecystectomy generally recommended due to ↑ risk of gallbladder cancer (mostly adenocarcinoma).

Acute cholangitis

Also called ascending cholangitis. Infection of biliary tree usually due to obstruction that leads to stasis/bacterial overgrowth.

Charcot triad of cholangitis includes jaundice, fever, RUQ pain.

Reynolds pentad is Charcot triad plus altered mental status and shock (hypotension).

Cholangiocarcinoma

Malignant tumor of bile duct epithelium. Most common location is convergence of right and left hepatic ducts. Risk factors include 1° sclerosing cholangitis, liver fluke infections (eg, *Clonorchis*). Usually presents late with fatigue, weight loss, abdominal pain, jaundice. Imaging may show biliary tract obstruction. Histology: infiltrating neoplastic glands associated with desmoplastic stroma.

Pancreatitis

Refers to inflammation of the pancreas. Usually sterile.

Acute pancreatitis

Autodigestion of pancreas by pancreatic enzymes (**A** shows pancreas [yellow arrows] surrounded by edema [red arrows]).

Causes: **I**diopathic, **G**allstones, **E**thanol, **T**rauma, **S**teroids, **M**umps, **A**utoimmune disease, **S**corpion sting, **H**ypercalcemia/**H**ypertriglyceridemia (> 1000 mg/dL), **E**RCP, **D**rugs (eg, sulfa drugs, NRTIs, protease inhibitors). **I GET SMASHED**.

Diagnosis by 2 of 3 criteria: acute epigastric pain often radiating to the back, serum amylase or lipase (more specific) to 3× upper limit of normal, or characteristic imaging findings.

Complications: pancreatic pseudocyst **B** (lined by granulation tissue, not epithelium), abscess, necrosis of parenchymal or peripancreatic tissue, hemorrhage, infection, organ failure (ALI/ARDS, shock, renal failure), hypocalcemia (precipitation of Ca^{2+} soaps).

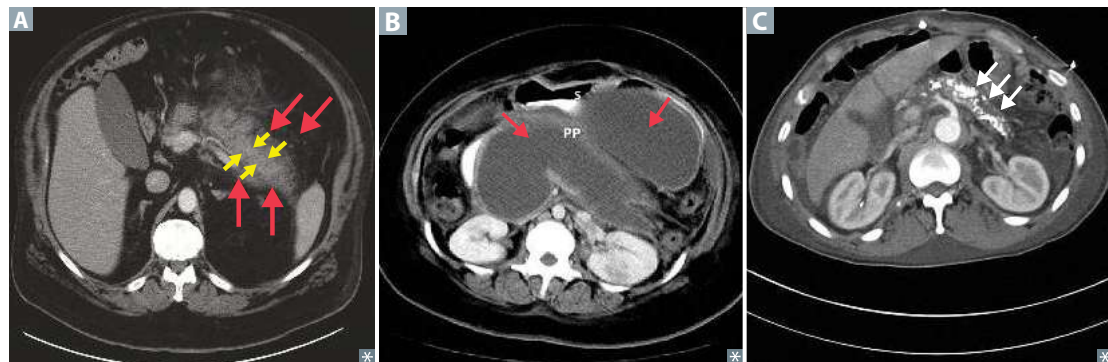
Chronic pancreatitis

Chronic inflammation, atrophy, calcification of the pancreas **C**. Major risk factors include alcohol use disorder and genetic predisposition (eg, cystic fibrosis, *SPINK1* mutations); can be idiopathic.

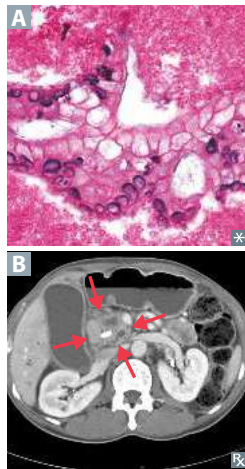
Complications include pancreatic insufficiency and pseudocysts.

Pancreatic insufficiency (typically when <10% pancreatic function) may manifest with steatorrhea, fat-soluble vitamin deficiency, diabetes mellitus.

Amylase and lipase may or may not be elevated (almost always elevated in acute pancreatitis).



Pancreatic adenocarcinoma



Very aggressive tumor arising from pancreatic ducts (disorganized glandular structure with cellular infiltration **A**); often metastatic at presentation, with average survival ~ 1 year after diagnosis. Tumors more common in pancreatic head **B** (lead to obstructive jaundice). Associated with CA 19-9 tumor marker (also CEA, less specific). Most common genomic abnormality is *KRAS*-activating mutation.

Risk factors:

- Tobacco smoking (strongest risk factor)
- Chronic pancreatitis (especially > 20 years)
- Diabetes
- Age > 50 years

Often presents with:

- Abdominal pain radiating to back
- Weight loss (due to malabsorption and anorexia)
- Migratory thrombophlebitis—redness and tenderness on palpation of extremities (Trousseau syndrome)
- Obstructive jaundice with palpable, nontender gallbladder (Courvoisier sign)

► GASTROINTESTINAL—PHARMACOLOGY

Acid suppression therapy

